AN-Najah National University

Department of Computer Engineering

VLSI (10636568) –HW#1

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Due: 3/11/2025

Solve this homework using Microwind and put all results in a **"pdf"** file and submit it. Also submit the layout file (the ***msk*** file)

Part1:

Consider the complex gate for the following Boolean Function: ***F = ABC+ D***

a. Draw the schematic and size the transistors such that the smallest transistor is 4/2. Make ***each stage*** is equivalent to the ***Unit Inverter***.
b. Draw the Layout using Microwind. Make sure you put the Body Contacts.
c. Draw a cross section in the **P** strip(s). Use the Snipping tool to extract the image)
d. Draw a cross section in the **N** strip(s). Use the Snipping tool to extract the image).
e. Put clocks at the Inputs A, B, C, D Simulate the Circuit and find the delays between the input and the output using the *built-in* SPICE Simulator. Show the delays (extract an Image using the Snipping Tool).

Important Notes:

1. Use the default setting for the Clocks. Make sure that d is the slowest. If you assign the clock in the Order A, B, C, D, it will automatically do it for you.
2. Your Circuit must produce the correct output. Verify its truth table.
3. Your circuit must satisfy the DRC rules; no DRC errors.
4. You must minimize the are as much as you can.

Part2:

Consider the complex gate for the following Boolean Function $\boldsymbol{F} = \overline{(\boldsymbol{A}+\boldsymbol{B})\boldsymbol{C}}$

a. Draw the schematic and size the transistors such that the smallest transistor is 4/2. Make itis equivalent to the ***Unit Inverter***.

b. Draw the stick diagram by hand (Not Microwind)
c. Use the Compile one line feature in Microwind and generate the layout. Include the generated layout in the pdf file for this assignment